

EMMEN AP, Switzerland

WMO: 066130

Lat: 47.084N

Lon: 8.296E

Elev: 427

StdP: 96.30

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.2	-7.2	-11.9	1.4	-8.8	-9.1	1.8	-5.8	8.3	8.0	7.2	3.8	1.9	30	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	30.2	20.7	28.3	19.8	26.3	18.9	21.4	28.6	20.3	26.7	19.5	25.3	2.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.4	25.1	18.0	13.6	23.8	17.1	12.9	22.6	64.1	28.7	60.5	27.0	57.6	25.5	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.3	4.5	-13.7	33.2	2.9	1.8	-15.7	34.4	-17.4	35.4	-19.0	36.4	-21.1	37.7		
(5)			-13.5	23.2	2.6	1.4	-15.4	24.2	-16.8	25.0	-18.3	25.8	-20.1	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	0.8	1.0	5.0	9.4	13.2	17.4	19.1	18.4	14.6	10.4	4.8	1.6	(6)
(7)		DBStd	7.53	4.21	4.06	3.64	3.80	3.79	3.35	3.17	3.07	3.04	3.48	3.71	3.50	(7)
(8)		HDD10.0	1228	286	253	156	57	14	1	0	0	2	39	160	259	(8)
(9)		HDD18.3	3286	544	487	412	269	164	56	28	38	116	247	407	518	(9)
(10)		CDD10.0	1112	1	0	2	37	112	223	283	259	141	51	3	0	(10)
(11)		CDD18.3	129	0	0	0	0	4	29	53	38	4	0	0	0	(11)
(12)		CDH23.3	1331	0	0	0	11	67	315	534	361	42	1	0	0	(12)
(13)		CDH26.7	377	0	0	0	0	12	88	177	96	4	0	0	0	(13)
(14)	Wind	WSAvg	1.4	1.4	1.4	1.6	1.6	1.7	1.6	1.6	1.3	1.2	1.2	1.1	1.3	(14)
(15)	Precipitation	PrecAvg	1509	82	90	95	116	166	175	175	124	103	106	103		(15)
(16)		PrecMax	1866	180	216	238	235	286	258	341	345	239	181	248	231	(16)
(17)		PrecMin	1174	10	36	34	20	39	94	70	44	45	9	1	2	(17)
(18)		PrecStd	173	46	45	47	54	65	44	68	69	47	41	54	55	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	13.9	19.1	24.9	28.1	32.0	32.9	32.0	27.0	22.1	16.9	11.9	(19)
MCWB			10.7	7.6	11.0	14.1	17.5	21.5	20.8	21.8	19.7	15.7	11.6	8.3	(20)	
2%		DB	11.1	11.2	16.9	22.8	25.0	29.1	31.0	29.2	24.2	19.8	14.2	9.9	(21)	
		MCWB	7.8	6.4	10.1	13.8	16.2	20.2	20.6	20.9	17.9	14.8	10.7	7.1	(22)	
5%		DB	8.8	9.0	14.8	20.2	23.0	27.0	28.9	27.0	22.8	18.0	12.0	8.2	(23)	
		MCWB	6.3	5.5	9.1	12.4	15.4	19.4	19.9	20.1	17.2	14.2	9.4	6.2	(24)	
10%		DB	6.3	6.9	12.2	17.9	20.9	24.8	26.2	25.1	20.9	16.0	10.2	6.9	(25)	
		MCWB	4.8	4.5	7.8	11.4	14.5	18.1	18.7	18.9	16.3	13.3	8.3	5.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	8.4	11.6	15.4	18.7	22.6	22.5	19.8	16.7	12.6	9.3	(27)	
MCDWB			14.4	12.3	17.7	22.8	26.9	30.3	30.8	30.5	25.6	20.4	16.0	11.1	(28)	
2%		WB	8.3	6.7	10.4	14.0	17.2	20.9	21.2	21.4	18.5	15.5	11.0	7.8	(29)	
		MCDWB	10.8	9.9	15.8	21.5	23.4	27.5	28.8	28.2	23.3	18.4	13.7	9.2	(30)	
5%		WB	6.4	5.6	9.4	12.9	16.0	19.7	20.1	20.2	17.5	14.5	9.8	6.5	(31)	
		MCDWB	8.0	8.6	14.1	19.6	21.2	25.7	27.0	26.0	21.8	17.2	11.7	7.7	(32)	
10%		WB	4.8	4.4	8.2	11.7	14.8	18.5	19.4	19.4	16.6	13.5	8.7	5.3	(33)	
		MCDWB	6.2	6.6	12.0	17.0	20.0	24.0	25.5	24.4	20.2	15.9	9.9	6.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.7	7.9	10.5	11.9	10.6	11.1	11.2	10.8	10.1	9.1	7.2	7.6	(35)
MCDBR			9.2	12.5	15.6	17.0	15.3	15.0	15.8	14.3	13.0	12.1	10.9	9.4	(36)	
MCWBR			6.3	8.2	9.5	9.4	7.5	7.2	6.8	7.1	7.4	7.6	7.5	6.7	(37)	
5% WB		MCDBR	8.6	11.3	14.4	16.2	13.4	13.9	14.2	13.6	12.1	10.5	9.7	8.6	(38)	
		MCWBR	6.2	7.8	9.0	9.1	6.8	7.0	6.7	7.0	6.9	6.9	7.0	6.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.294	0.324	0.370	0.410	0.427	0.432	0.434	0.422	0.392	0.364	0.324	0.291	(40)	
taud		2.369	2.289	2.231	2.183	2.194	2.234	2.228	2.273	2.335	2.386	2.425	2.391	(41)		
Ebn at Noon		792	834	843	842	840	836	828	822	813	780	749	750	(42)		
Edh at Noon		71	95	118	136	140	135	134	124	106	87	68	62	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.31	2.19	3.40	4.38	4.93	5.74	5.60	4.88	3.72	2.39	1.37	1.08	(44)	
RadStd		0.15	0.29	0.48	0.66	0.49	0.53	0.50	0.47	0.37	0.26	0.18	0.16	(45)		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (120 neighbors)		N/A	N/A	N/A	+0.71	+0.41	+0.32	N/A	N/A	+57	+15

Nomenclature: See separate page